

19 Genetic technology

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

PCR cycles denature DNA, anneal primers and extend strands with thermostable polymerase; product can double approximately each cycle under ideal conditions. Restriction enzymes cut defined sequences, ligase joins DNA, and vectors introduce constructs into cells. Gel electrophoresis separates fragments by size; controls are needed to interpret band patterns and limit contamination.

Worked example: follow the reasoning

Starting with one target copy, ideal PCR produces $2^5 = 32$ copies after five complete cycles; actual yields may be lower.

Quick check · multiple choice

A Which enzyme extends primers in PCR?

[1]

- A DNA ligase
- B Restriction endonuclease
- C Thermostable DNA polymerase
- D RNA polymerase only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A PCR sample starts with 10 double-stranded targets. Estimate ideal copies after six cycles and give one reason actual yield differs.

[3]

Final answer: _____

Answers and explanation

Quick check: C. Thermostable DNA polymerase

Question 1 [3 marks]

$10 \times 2^6 = 640$ copies [2]; reagent depletion, incomplete efficiency or inhibitors reduce actual yield [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.